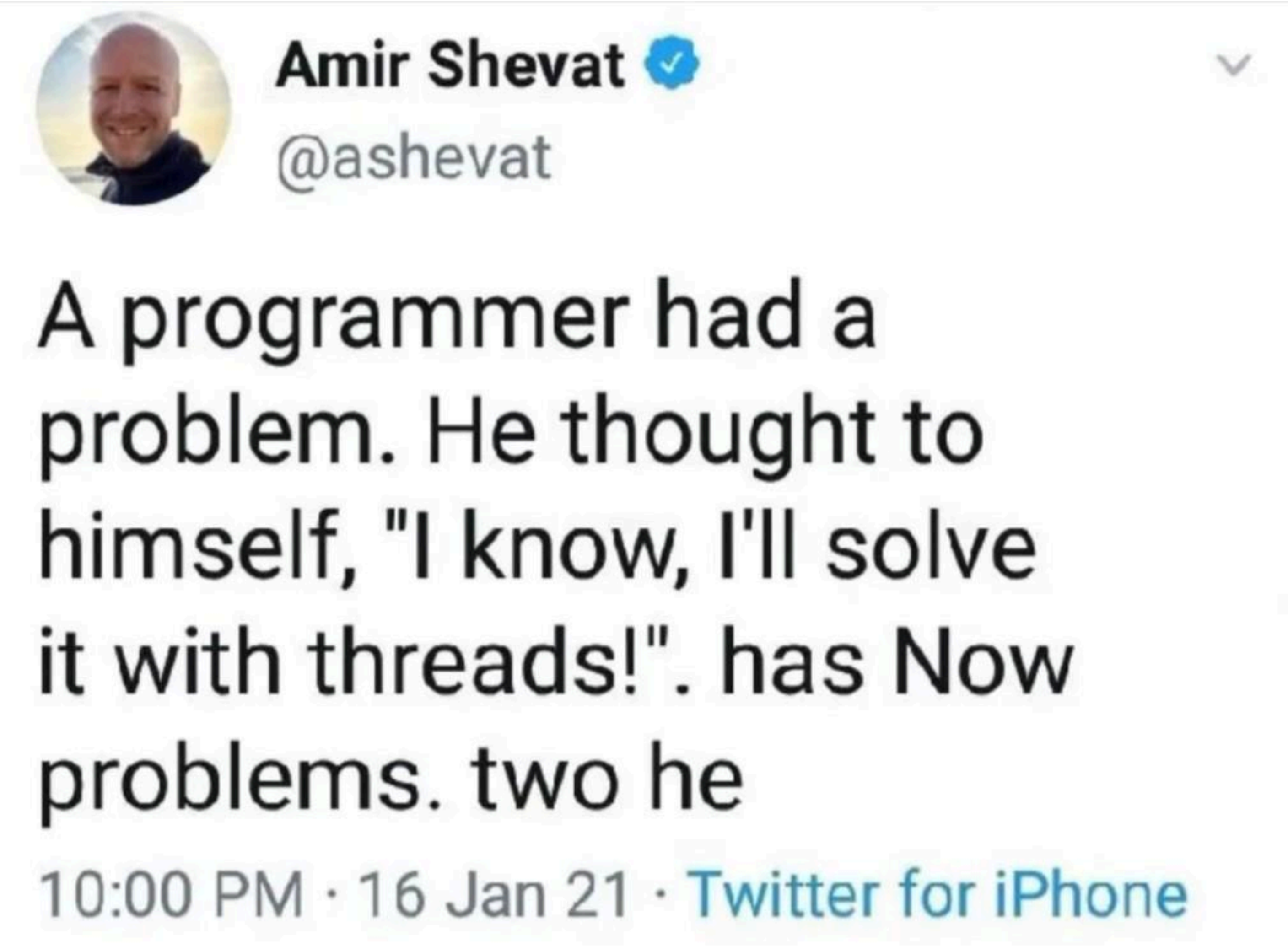
Eraser notes
(240lx spr26)



Eraser: Savage, Burrows et al.

Famous.

- Only one of 3 non-performance SOSP papers in the 90s.
- Catches race conditions by looking for inconsistent locks.
- Uses dynamic binary translation (ATOM) to check a "consistent" lock is held for each load/store of memory word.

Problem:

- DBT = 1MLOC system (ATOM).
- Notable if can run in kernel.

Today:

- use your mem trap to make Eraser in a < 300 lines.

Basic Eraser idea: lockset

Definitions:

- th.lockset: the locks a thread holds now.
- mem.lockset: intersection of locks held when mem accessed in past.

Transition table:

- th=fork(...): th.lockset = {}
- mem=malloc(): mem.lockset = {all}
- lock(l1): th.lockset U= {l1}
- unlock(l1): th.lockset -= {l1}
- load [mem], or store [mem]:
    - mem.lockset = th.lockset \inter mem.lockset
    - if mem.lockset = {} error

Example

```
     x.lockset={all}
     T1.lockset = {}
-----------------------------------
    T1          |
   lock(L1)     |   T1.lockset = {L1}
   x++;         |   x.lockset = {L1}
   unlock(L1)   |   T1.lockset = {}
        <switchto T2>
    T2
   lock(L2)     |   T2.lockset = {L2}
   x++;         |   x.lockset = {}
                    error "empty lockset for x!"
```

Key:

- order of set intersection doesn't matter.
- so find error no matter what order we run T1,T2!

Hack 1: initialize mem without locks.

Common pattern:

- initialize mem without holding mem's lock.
- This is fine b/c mem not reachable yet.
- Not reachable = no concurrent access possible.
- No lock needed.

We can't compute reachable(mem), so do a hack:

- track which thread accesses mem first.
- don't track its lockset. no errors.
- when *another* thread T access mem, then mem.lockset = T.lockset.
- track it as usual.

Hack 2: initialize mem, then read-only.

Common pattern:

- initialize mem with stores.
- release it.
- all threads access read-only.
- Don't need locks b/c no mutation.

Hack:

- Do the initialization hack, but as long as accesses are loads, don't flag.
- If there is a store: flag as normal.

Today

- We give you a trivial eraser that only tracks one lock and one memory location.
- Make it track multiple of both (write some tests).
- Add the two hacks above
- Tons of extensions: doing shadow memory is great!

Note:

- Very reasonable lab for Daniel mode.
- You'll have to update the readme halfway through --- I'm moving some tests around.